This WACV paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

CLIP-UP: CLIP-Based Unanswerable Problem Detection for Visual Question Answering

Ben Vardi[1] Oron Nir[1,2] Ariel Shamir[1]
[1]Reichman University [2]Microsoft Corporation
https://benvr.github.io/CLIP-UP

Abstract

Vision-Language Models (VLMs) demonstrate remarkable capabilities in visual understanding and reasoning, such as in Visual Question Answering (VQA), where the model is asked a question related to a visual input. Still, these models can make distinctly unnatural errors, for example, providing (wrong) answers to unanswerable VQA questions, such as questions asking about objects that do not appear in the image.

To address this issue, we propose CLIP-UP: CLIP-based Unanswerable Problem detection, a novel lightweight method for equipping VLMs with the ability to withhold answers to unanswerable questions. CLIP-UP leverages CLIP-based similarity measures to extract question-image alignment information to detect unanswerability, requiring efficient training of only a few additional layers, while keeping the original VLMs' weights unchanged.

Tested across several models, CLIP-UP achieves significant improvements on benchmarks assessing unanswerability in both multiple-choice and open-ended VQA, surpassing other methods, while preserving original performance on other tasks.

1. Introduction

A fundamental task in the domain of Vision-Language Models (VLMs) is Visual Question Answering (VQA) [3], where the model is asked a question related to a visual input. VQA appears in various formats, with open-ended questions being the most immediate form. Multiple-choice VQA offers a more constrained setup, requiring to discriminate between several plausible answer options.

Recent VLMs [37, 42, 67] excel in both VQA formats but a critical challenge persists: the tendency of VLMs to produce incorrect or irrelevant responses, a phenomenon commonly referred to as "hallucinations" [38, 52]. The problem is particularly concerning in VQA, where models might confidently provide answers that, while seemingly plausible, are actually inconsistent with the visual content [33, 45]. This raises concerns about VLMs' reliability and applicability, indicating the need for mitigating such hallucinations [5].

In this work, we focus on a crucial hallucination challenge within VLMs: their tendency to provide answers to unanswerable visual questions, flawed either in the question-image pairing or the question itself, in both multiple-choice and open-ended VQA. For multiple-choice VQA, this issue was formalized as the *Unsolvable Problem Detection (UPD)* challenge [45]. In this challenge, models are evaluated first on their ability to detect unanswerable VQA inputs and withhold answers when necessary, and second on correctly answering answerable questions. Hence, this task is more complex than a simple binary classification of unanswerability. Visual unanswerable multiple-choice questions are classified into three categories [45]: (1) Absent Answer Detection (AAD): detecting questions where all answer options are incorrect; (2) Incompatible Answer Set Detection (IASD): detecting questions with answer options incompatible with the question; and (3) Incompatible Visual Question Detection (IVQD): detecting questions where the question is incompatible with the image (see Fig. 1 for illustration).

The importance of UPD in multiple-choice VQA lies not only in improving model robustness but also in the fact that multiple-choice questions are commonly used to benchmark VLMs, as the structured format simplifies evaluation compared to open-ended questions. However, this format can obscure the actual understanding of the model. For example, rather than genuinely reasoning about the input, models may rely on shortcut strategies, such as eliminating unlikely options. Equipping models with the ability to withhold answers to unanswerable questions discourages such behavior and enables more reliable evaluations that better reflect models' true understanding.

For open-ended VQA, the importance of UPD lies in ensuring that models are robust and capable of handling unanswerable questions that arise in real-world scenarios, such as when assisting visually impaired individuals [20].

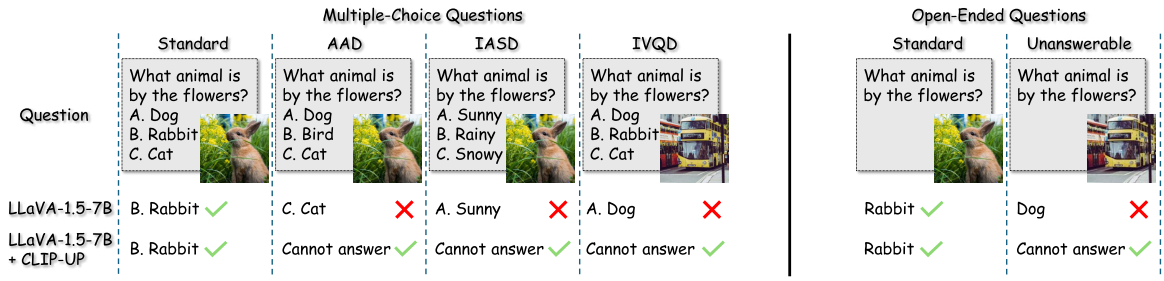


Figure 1. CLIP-UP equips VLMs such as LLaVA-1.5-7B [36] with the ability to detect and withhold answers to multiple-choice and open-ended unanswerable questions, while preserving models' original capabilities on standard answerable questions.

An examination of VLMs' training data reveals a predominance of valid, answerable questions [36], likely contributing to their tendency to always provide an answer. A straightforward solution to UPD is therefore to fine-tune or re-train the full model with data that include unanswerable questions. However, this approach is often impractical due to the high computational cost and data requirements. Therefore, a pressing question arises: how can we adapt existing models to identify when they should refrain from answering in affordable costs?

To tackle this question, Miyai et al. [45] and Qian et al. [50] explored prompt engineering solutions that modify the input to inform the model it may withhold an answer. This is done, for example, by adding a "None of the above" option in multiple-choice questions, providing an alternative when the question is unanswerable. While appealing due to its training-free nature and ease of implementation, prompt engineering was found to provide only limited improvement. Another solution is to fine-tune models on VQA-specific data, including only answerable and unanswerable questions [8, 45]. Although this approach significantly improves models' ability to detect unanswerable questions, its exclusive focus on VQA compromises performance on tasks other than VQA.

In this work, we introduce CLIP-based Unanswerable Problem detection (CLIP-UP): a lightweight method for enhancing general pre-trained VLMs with the capability to detect unanswerable questions in both multiple-choice and open-ended VQA formats (see Fig. 1). CLIP-UP operates by leveraging carefully crafted *correlation vectors* derived from CLIP embeddings [51] of the input image and question, encoding image-question alignment information. These vectors are projected into the VLM's intermediate feature space, producing a new embedding vector that serves as an answerability prior and is seamlessly integrated into the model. CLIP-UP this way trains only a single linear projection layer for each question format (multiple-choice or open-ended) to create this vector, while keeping the original VLM weights unchanged.

Using a simple classifier, we determine whether and which embedding vector should be activated: for multiple-choice or open-ended VQA inputs, the corresponding vector is generated and integrated to enhance UPD capabilities; otherwise, no vector is generated, ensuring the model's original capabilities remain intact on non-VQA tasks.

The effectiveness of CLIP-UP depends on the quality of the CLIP signal and the use of this information. In particular, we use Structure-CLIP [25], a CLIP variant that offers improved sensitivity for texts with similar structures but different semantics. Beyond using correlation vectors to generate an embedding vector fed into the VLM, we also propose *Injected LoRA*, a novel approach for injecting priors directly into LoRA [23] layers, where the prior in our case is the correlation vectors.

Our experiments on multiple VQA unanswerability benchmarks show that the UPD problem persists in recent VLMs and that CLIP-UP, applied across different models, achieves performance surpassing other UPD enhancement methods. We also created a new multiple-choice dataset for training CLIP-UP, covering all unanswerability categories.

Our contributions can be summarized as follows:

- We introduce CLIP-UP, a novel lightweight approach leveraging CLIP to enhance VLMs' ability to withhold answers to unanswerable questions, while leaving the original VLM weights unaltered.
- We demonstrate that CLIP-UP significantly improves UPD performance in multiple-choice and open-ended VQA across various models, outperforming other methods while preserving performance on non-VQA tasks.
- We propose a new method for injecting priors into LoRA layers, and show that this way of using CLIP-UP correlation vectors improves results over standard LoRA.
- We release our code and multiple-choice UPD training dataset to support future research.

2. Related work

Vision-language models The rapid advancements in Large Language Models (LLMs) in recent years [6, 11, 12, 59] have led to impressive performance across a wide

range of text-based tasks. Building on the success of LLMs, Vision-Language Models (VLMs) have emerged by integrating visual inputs into LLMs, enabling models to reason about images and text simultaneously [1, 35, 49, 67].

VLMs typically process images through a pre-trained vision encoder, creating embeddings that are subsequently aligned with and fed into a pre-trained LLM.

Hallucination issues in VLMs Despite their significant progress, VLMs often generate text that is semantically coherent but conflicts with the content of the input image, a problem generally referred to as "hallucinations" [38, 52]. Causes of VLMs' hallucinations are diverse, including biases in fine-tuning data [33], image encoder limitations [15, 58], and language biases in the LLM decoder [16, 46]. To address hallucinations, various mitigation methods and datasets [15, 17, 18, 33, 57, 58] have been proposed.

Although hallucination mitigation is a popular research focus, it remains a key challenge for VLMs. In this work, we focus on a specific hallucination category, where models do not withhold answers to unanswerable visual questions.

Unanswerable visual question answering In Visual Question Answering (VQA) [3], VLMs are shown an image and a related question and are expected to provide an accurate response. While early benchmarks focused on open-ended questions [3, 16, 30], which is the most elementary and real-world version, more recent works include other VQA types, such as yes/no [13] and multiple-choice questions [31, 39, 63]. Among these, multiple-choice VQA, offering a closed set of options, has become an important VQA variant and a primary testbed for assessing VLMs, with models [1, 35, 49] rigorously tested on its benchmarks.

However, most multiple-choice and open-ended VQA benchmarks contain only answerable questions. For the open-ended setting, this means that the question and image are compatible. For multiple-choice, it means that the question, image, and answer set are all compatible, and a correct answer exists within the set. This setup does not reflect real-world scenarios, where questions may be unanswerable.

To address this gap in multiple-choice VQA, Miyai et al. [45] introduced the Unsolvable Problem Detection (UPD) challenge, formalizing unanswerability in the multiple-choice setting. They defined three unanswerability types (AAD, IASD, IVQD; see explanation in Sec. 1) and published the MM-UPD benchmark, containing flawed questions alongside standard answerable questions. VQA unanswerability here and in our work refers to clear flaws in the VQA input, and not knowledge gaps or uncertainty [8, 22]. Using the MM-UPD benchmark, Miyai et al. [45] showed that VLMs often respond with an answer even when no relevant option exists, with open-source models showing particularly low performance. A similar finding was reported for finer-grained multiple-choice answerability types in [22].

For open-ended VQA, no prior work has formalized coarse types of unanswerability, as it appears that only one type exists, namely the incompatibility between the image and the question (parallel to IVQD in multiple-choice VQA) [19, 50, 66]. However, some works [8, 20, 50] have proposed finer-grained categories of open-ended unanswerability, such as visually deceptive images [50] or ambiguity due to poor image quality [20]. Similar to the multiple-choice case, popular VLMs have been found to perform poorly on unanswerable open-ended questions [8, 50].

In this work, we show that although improved in recent VLMs, these challenges persist and present a solution leveraging CLIP in a lightweight training framework.

Efficient model editing Model editing is a research area focused on making targeted changes to pre-trained models for specific inputs, without compromising overall performance. Efficiency is a key goal, with efforts aimed at developing methods to apply edits without high computational or extensive data requirements [62, 65].

Model editing has gained attention in the context of LLMs, driven by the growing need to adapt models to issues such as updating outdated information [7, 21, 44]. It is also popular in text-to-image generation, where personalization may be achieved by affordable learning of concept-specific embedding vectors [4, 14, 53, 60].

Few works have explored model editing in VLMs [2, 10, 24]. MyVLM [2], for instance, learns concept-specific embedding vectors for personalized VLM outputs. Our CLIP-UP method also introduces a new embedding vector, but to enhance VLMs with the ability to withhold answers to unanswerable questions. Moreover, rather than learning the vector from scratch, we leverage CLIP's vision-language alignment and learn projection layers to create it.

3. Method

This section first describes CLIP-UP embedding injection for multiple-choice questions, and then explains how it extends to open-ended questions with minimal changes. An overview of our approach applied to common VLM architectures is shown in Fig. 2a. Given an image and a multiple-choice VQA prompt, we first parse the text to form individual text segments that merge the question with each answer. Each text segment is encoded by CLIP-based text encoder, while the image is encoded by CLIP-based image encoder. Each text segment encoding is multiplied element-wise by the image encoding to create *correlation vectors*. These vectors capture the alignment between the image and each answer option (see Sec. 3.1). Next, we concatenate the correlation vectors and pass them through a learnable projection layer, transforming them into a new embedding vector

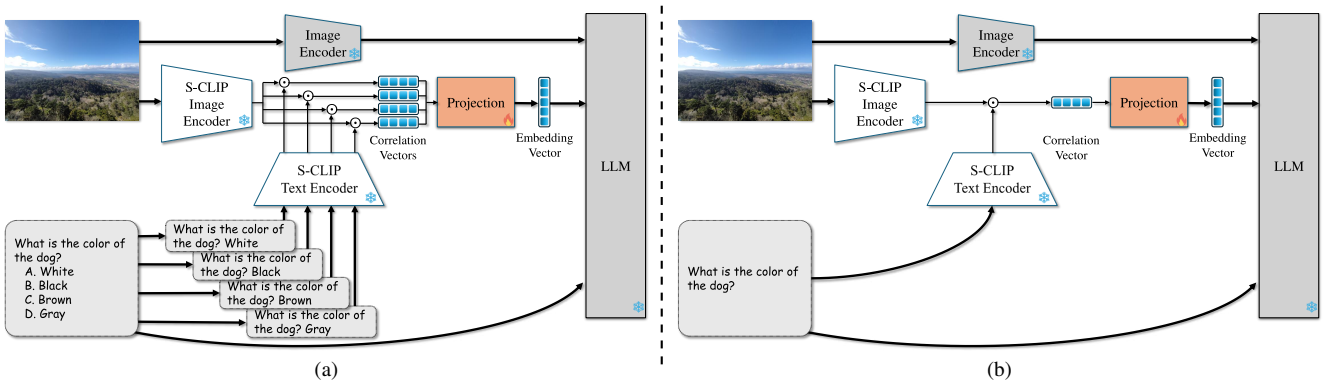


Figure 2. CLIP-UP embedding injection applied on common VLM architectures. (a) For multiple-choice questions, given an image and a VQA prompt, the prompt is transformed into text segments merging the question with each answer option. These segments and the image are encoded by Structure-CLIP (S-CLIP) to produce embeddings, from which correlation vectors are formed via element-wise multiplication. A learnable projection module maps these vectors into the VLM's intermediate feature space. The resulting new embedding vector is integrated into the LLM component of the VLM alongside the standard inputs. (b) For open-ended questions, the process is similar but involves a single correlation vector computed from the image and question.

within the VLM's feature space. This embedding is then fed into the LLM component of the VLM, alongside standard inputs (see Sec. 3.2).

CLIP-UP for open-ended questions has one key difference: instead of computing multiple correlation vectors by correlating the image with answer options, it computes a single correlation vector between the image and the question (see Sec. 3.3 and Fig. 2b).

Finally, we explore alternative ways of injecting the correlation vectors into the VLM. Specifically, we propose a novel method that injects the correlation vectors directly into LoRA layers during LoRA fine-tuning (see Sec. 3.4).

3.1. Correlation vectors generation

The first step of CLIP-UP for multiple-choice VQA is to generate correlation vectors. Let (T, I) be a multiple-choice VQA input, where T represents the text (both the question and answer options) and I is the image. Given such input with n options, we parse T into a pair $(Q, \{O_1, \ldots, O_n\})$, where Q is the question and $\{O_1, \ldots, O_n\}$ is the set of answer options. Parsing is done with a simple rule-based algorithm; see App. D in the Supplementary Material (Supp.) for details.

To contextualize each answer option O_i, we merge it with the question Q. This forms the set $\mathcal{Q}_{\text{opt}} = \{Q + O_i \mid i = 1, \ldots, n\}$, where each $s_i \in \mathcal{Q}_{\text{opt}}$ is the question followed by a single answer option. Note that using answers alone does not contain sufficient information, as it can lack context (*e.g.*, "Blue" vs. "What color is the dress? Blue").

We then encode each contextualized answer option $s_i \in \mathcal{Q}_{\text{opt}}$ with Structure-CLIP [25] text encoder to obtain a text embedding $\mathbf{v}_{s_i} = \text{SC}_{\text{text}}(s_i)$. The image is processed by Structure-CLIP image encoder, yielding the image embedding $\mathbf{v}_I = \text{SC}_{\text{img}}(I)$. Structure-CLIP [25] is a variant of CLIP, designed to better distinguish semantically different texts with similar structures. It is well-suited for our contextualized answer options which share the same structure, namely, the same question followed by an answer option.

We generate n correlation vectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ by performing element-wise multiplication between each text embedding and the image embedding:

$$\mathbf{u}_i = \mathbf{v}_I \odot \mathbf{v}_{s_i}. \tag{1}$$

Rather than relying solely on scalar CLIP-based similarity scores (dot product of embeddings), our approach constructs richer correlation vectors incorporating similarity scores (as the sum of an element-wise product is the dot product) and additional alignment information.

Extracting these correlation vectors provides a strong prior for assessing the answerability of the multiple-choice VQA input. As illustrated in Figs. 3a–3d, for a standard answerable question (*i.e.*, one having a correct answer option) paired with its correct answer, the image and text align well, leading to high CLIP similarity. Consequently, for standard questions, one correlation vector, the one corresponding to the correct answer, will exhibit high values. Conversely, for unanswerable questions, all n correlation vectors are expected to exhibit low values, as no contextualized option aligns well with the image. Therefore the n correlation vectors provide a strong signal for determining multiple-choice VQA answerability.

3.2. Learning a new projection layer

The correlation vectors capture essential alignment signal, but the VLM cannot directly interpret them, as they are neither optimized for its use nor aligned with its feature space dimension. To address this, we concatenate the correlation

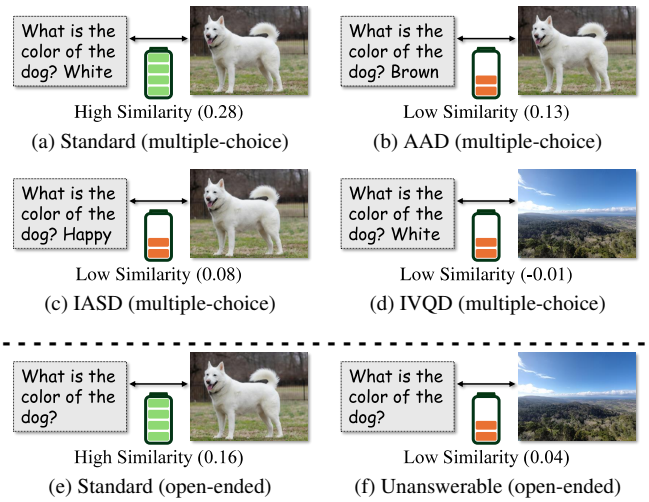


Figure 3. Our correlation vectors capture a prior for VQA answerability. In multiple-choice questions, (a) for a standard question, the correct contextualized answer option aligns well with the image, resulting in a correlation vector with high values. For unanswerable questions, no option aligns well with the image: either (b) all answer options are incorrect (AAD), (c) incorrect and irrelevant to the question (IASD), or (d) the question is incompatible with the image (IVQD). Open-ended questions show a similar trend: (e) answerable questions align well with the image, in contrast to (f) unanswerable ones. Numbers represent average Structure-CLIP similarity scores measured on test data.

vectors and project them into a vector $\mathbf{e}$ in the VLM's feature space, using a learnable projection layer $\mathcal{P}$:

$$\mathbf{e} = \mathcal{P}(\mathbf{u}), \text{ where } \mathbf{u} = [\mathbf{u}_1; \ldots; \mathbf{u}_n]. \tag{2}$$

The new embedding vector $\mathbf{e}$ is subsequently fed into the LLM component of the VLM. In particular, only the projection layer is trained, while all other components remain frozen, making CLIP-UP training simple and efficient. Training uses cross-entropy loss on answerable and unanswerable questions: answerable questions have the correct answer option (*e.g.*, "A. White") as their ground truth text, while unanswerable questions use "I cannot answer."

We use a simple linear layer for the projection. Since the projection input size is fixed, the concatenated correlation vectors must have fixed size. Given that multiple-choice questions typically have up to four options, we generate four correlation vectors for each input. For inputs with fewer options (three or two), we fill the remaining slots with correlation vectors generated from element-wise multiplication of the image embedding and null text embedding.

What makes CLIP-UP effective at enhancing UPD performance, even though VLMs already "see" both the image and text? We postulate that the global alignment information extracted by CLIP-UP is absent in popular VLMs [1, 32, 35–37, 67]. For example, although LLaVA uses a CLIP image encoder, it extracts patch features from the penultimate layer, assumed to be more effective for capturing image details [35]. InternVL3 [67] uses an earlier layer of its InternViT2.5 vision encoder [9] (layer 45 out of 48). In contrast, CLIP-UP equips VLMs with global information by using Structure-CLIP's class embedding from the last layer, explicitly trained to capture global alignment [51]. Together with the incorporation of Structure-CLIP's text embeddings, CLIP-UP introduces global image-text information that VLMs lack.

To recap, our goal is to create a new embedding vector that conveys (un)answerability information to the model. Fig. 4 illustrates this intuition, showing that the learned projection forms distinct clusters of the embeddings of "answerable" and "unanswerable" vectors.

3.3. Handling open-ended questions

The CLIP-UP scheme proposed for multiple-choice VQA can be applied to open-ended questions. The key difference is that in the absence of answer options, the method computes a single correlation vector between the image and the question, expected to exhibit higher values for answerable questions (see Figs. 3e–3f). This vector is then projected using a learnable projection layer, trained on both answerable and unanswerable open-ended questions (see Fig. 2b).

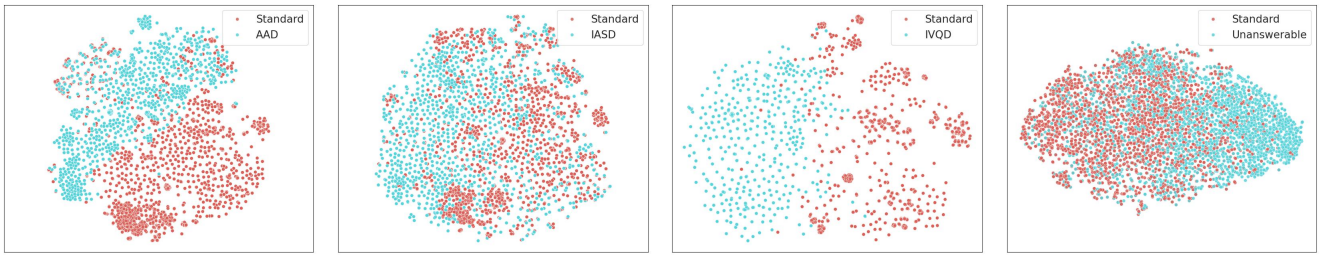
Although CLIP-UP generates separate embedding vectors for multiple-choice and open-ended questions, we can flexibly activate either one or none. During inference, a simple classifier determines whether the input is a multiple-choice question, an open-ended question, or neither. This enables generating and using the relevant embedding vector, or omitting it entirely if the input is not a question. Since the original VLM weights are preserved, the model performance on non-VQA tasks remains unaffected.

3.4. Injecting correlation vectors into LoRA

The core idea of CLIP-UP is to inject an answerability prior into the VLM's decision process. So far, this has been achieved by injecting a new embedding vector into the VLM directly. However, other injection approaches are possible. In particular, we propose a general method to inject information directly into LoRA layers during LoRA fine-tuning [23], referred to as *Injected LoRA (InjLoRA)*, and use it to inject our correlation vectors. InjLoRA provides a more direct and expressive injection mechanism to modulate LoRA weights than prior work. For example, Stracke et al. [56] inject signals via shift and scale operations.

In standard LoRA fine-tuning, pretrained weights $W_0 \in \mathbb{R}^{d\times k}$ are modified to $W_0 + \Delta W$, where $\Delta W = BA$, with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and r denoting the LoRA rank. In our InjLoRA, we learn a projection layer $\mathcal{P}'$ for each LoRA layer, such that ΔW becomes:

$$\Delta W = B\left(\mathcal{P}'(\mathbf{u}) + C\right)A, \tag{3}$$



(a) AAD (b) IASD (c) IVQD (d) Open-ended

Figure 4. t-SNE plots of embedding vectors generated by CLIP-UP on LLaVA-1.5-7B, for (a-c) all samples in MM-UPD, and (d) all samples in the RGQA test data.

where $\mathbf{u}$ is the concatenated correlation vectors (from Eq. (2)), and $\mathcal{P}'(\mathbf{u}) \in \mathbb{R}^{r\times r}$. The matrix $C \in \mathbb{R}^{r\times r}$ is learnable and acts as a residual, allowing LoRA layers to ignore the injected signal if desired.

We learn a different projection $\mathcal{P}'$ for each LoRA layer, allowing each layer to attend to different aspects of the correlation signal according to its role in the model.

4. Experiments

4.1. Datasets

Multiple-choice VQA We created a multiple-choice VQA training dataset, containing answerable and unanswerable questions across AAD, IASD, and IVQD. The dataset is organized into question pairs, each consists of a standard question and its corresponding unanswerable variant. For example, an AAD unanswerable question is generated by removing the correct option from a standard question. The dataset includes 293, 189, and 307 question pairs for AAD, IASD, and IVQD, respectively. For each category, 30 pairs are allocated for validation, and the rest used for training. The dataset will be released. See more details in App. B.1.

Open-ended VQA For open-ended VQA training, we use 700 answerable-unanswerable question pairs sampled from the TDIUC training dataset [27], with 50 pairs assigned for validation and the rest for training.

4.2. Training and models

We evaluate two CLIP-UP methods: (1) CLIP-UP with embedding injection only (*CLIP-UP-Emb*), and (2) CLIP-UP with both embedding injection and InjLoRA fine-tuning (*CLIP-UP-EmbLoRA*). To avoid confusion, we use the term CLIP-UP hereafter to refer to the general framework, while CLIP-UP-Emb and CLIP-UP-EmbLoRA denote the specific implementations. Note that although we evaluate InjLoRA alongside embedding injection (in CLIP-UP-EmbLoRA), injection into LoRA layers via InjLoRA can also be used independently.

Experiments are conducted on six VLMs with diverse architectures, scales, and initial UPD ability: LLaVA-1.5-7B [36], LLaVA-NeXT-13B [37], Phi-3.5-Vision [1], Ovis2-16B [42], InternVL3-1B, and InternVL3-8B [67].

We train CLIP-UP separately on multiple-choice and open-ended data. In both cases, training is conducted for 3 epochs with a cross-entropy loss. We train with batches containing pairs of answerable and unanswerable questions (*e.g.*, batch size of 8 with 4 pairs), which we found to improve training stability. See App. A for additional details.

4.3. Method comparisons

Multiple-choice VQA We first compare CLIP-UP to the original VLMs, where multiple-choice VQA is evaluated with the instruction "Answer directly with the letter of the correct option from the given choices" added to the prompt [45]. We also compare to prior UPD solutions, including prompt engineering settings from [45]: Base Prompt, with no added instructions, and Additional-Option and Additional-Instruction which encourage VLMs to withhold answers to unanswerable questions. For example, Additional-Option adds an option indicating that no answer is correct (see App. F.2). Finally, we compare CLIP-UP to LoRA fine-tuning [23] following [45]. To ensure strict evaluation, we use each VLM's recommended LoRA settings, which are more expressive (up to rank 128) than those used in our multiple-choice experiments with InjLoRA (rank 8).

Open-ended VQA For open-ended questions, we first compare to the original models, evaluated using the instruction "Answer the question using a single word or phrase" as in [36]. We also test prompt engineering by adding "When the provided information is insufficient, respond with 'I cannot answer'." to the original prompt. Finally, we compare CLIP-UP methods to LoRA fine-tuning, as it is the most effective baseline for multiple-choice questions.

4.4. Benchmarks and evaluation metrics

Multiple-choice experiments We evaluate CLIP-UP on the comprehensive MM-UPD benchmark [45], which con-

Method	LLaVA-1.5-7B			Phi-3.5-Vision			InternVL3-8B			Ovis2-16B		
	Stand.	UPD	Dual	Stand.	UPD	Dual	Stand.	UPD	Dual	Stand.	UPD	Dual
Original Model	67.73	0.11	0.11	78.58	0.41	0.41	88.95	0.00	0.00	88.38	9.97	9.65
Base Setting	66.33	7.23	4.87	77.12	0.92	0.89	86.32	44.46	39.45	87.03	46.46	42.44
Additional-Option	65.96	51.25	38.11	77.52	39.77	32.73	88.81	66.50	60.06	87.52	70.89	63.57
Additional-Instruction	65.88	43.61	31.39	75.80	50.32	38.99	86.44	80.15	70.44	87.01	80.15	71.69
LoRA Fine-Tuning	62.60	76.70	50.25	60.02	86.99	53.69	86.28	82.10	71.59	85.88	87.62	76.65
CLIP-UP-Emb (ours)	59.72	80.30	51.22	60.96	87.82	54.07	84.09	90.98	77.62	83.50	91.63	77.92
CLIP-UP-EmbLoRA (ours)	60.72	83.34	**53.07**	70.17	86.62	**62.27**	85.83	90.88	**79.22**	87.48	90.20	**80.59**

Table 1. Results (%) on MM-UPD [45] multiple-choice VQA. Metrics include circular standard, UPD, and dual accuracies.

Method	LLaVA-1.5-7B			Phi-3.5-Vision			InternVL3-8B			Ovis2-16B		
	Stand.	UPD	Dual	Stand.	UPD	Dual	Stand.	UPD	Dual	Stand.	UPD	Dual
Original Model	68.40	11.08	7.22	69.76	9.20	6.72	69.84	14.80	10.88	69.56	12.18	8.68
Prompt Engineering	67.24	14.52	9.56	69.30	25.44	18.06	67.26	55.22	35.90	29.44	94.40	26.58
LoRA Fine-Tuning	57.44	63.14	31.78	68.66	47.60	32.20	67.14	61.74	39.84	64.62	68.18	41.98
CLIP-UP-Emb (ours)	39.74	77.14	28.42	60.52	60.00	32.96	67.04	57.58	38.42	61.46	70.46	41.56
CLIP-UP-EmbLoRA (ours)	57.64	60.64	**32.36**	67.74	61.26	**40.28**	66.06	67.04	**42.88**	62.58	74.10	**44.52**

Table 2. Results (%) on RGQA [66] open-ended VQA. Metrics include standard, UPD, and dual accuracies.

sists of three sub-benchmarks for AAD, IASD, and IVQD. Each sub-benchmark contains pairs of multiple-choice VQA questions, an answerable one and an unanswerable one. To account for options' order variation, the questions are repeated n times (n is the number of options), with a different circular shift applied to the options each time.

Evaluation uses three metrics: circular standard accuracy, circular UPD accuracy, and circular dual accuracy. Circular accuracy [39] evaluates accuracy across all circular shifts of options by counting success only if all shifts are answered correctly. In particular, circular standard and UPD accuracies measure the circular accuracy for answerable and unanswerable questions, respectively.

Circular dual accuracy [45] requires correctly answering all circular variants of both standard questions and their unanswerable pairs. We use it as our main metric, as high dual accuracy indicates consistent discernment of answerability, in addition to correctly answering answerable questions. Note that, in contrast, maximizing either standard or UPD accuracy individually is trivial (by using the VLM as-is or by always refraining from answering, respectively).

Open-ended experiments We evaluate open-ended VQA on 5,000 examples from each of the RGQA benchmark [66] and TDIUC test set [27], each consisting of answerable-unanswerable question pairs. We report standard, UPD, and dual accuracies, all measured in regular form. Evaluation follows the LAVE metric [43], which leverages LLMs for open-ended VQA scoring.

4.5. Results

Multiple-choice results Tab. 1 presents the MM-UPD results of CLIP-UP methods applied to LLaVA-1.5-7B, Phi-3.5-Vision, InternVL3-8B, and Ovis2-16B averaged over AAD, IASD, and IVQD. Tabs. 5 and 6 (in Supp.) show the full results, including those for other VLMs.

Although recent VLMs have improved (*e.g.* Ovis2-16B), large gaps remain between standard and dual accuracies. CLIP-UP-EmbLoRA addresses this, achieving the best dual accuracy with gains up to 8.58% over LoRA fine-tuning. For example, it improves performance by 7.63% on the strong InternVL3-8B, demonstrating that it can effectively address the UPD problem. CLIP-UP-Emb also achieves strong performance, surpassing all baselines in most cases, further showing the effectiveness of the prior injection.

Although CLIP-UP methods improve UPD performance, the lightweight training is not intended to boost standard accuracy. The standard accuracy of the original model therefore serves as an upper bound for dual accuracy. Thus, CLIP-UP is closer to its upper limit than it may seem. For example, on Ovis2-16B, CLIP-UP-EmbLoRA achieves 80.59% dual accuracy, where the upper bound is 88.38%.

However, this upper bound may not reflect the true knowledge of the model, as the original model may rely on answer elimination shortcut strategy. Since CLIP-UP discourages this strategy, the standard accuracy with CLIP-UP may better reflect the model's actual understanding.

Improving dual accuracy comes at the cost of reduced standard accuracy, reflecting a trade-off. In some cases,

Method	Stand.	UPD	Dual
CLIP-UP-Emb w/ const. signal	56.82	75.17	45.27
CLIP-UP-Emb w/ similarities	60.39	73.84	47.90
CLIP-UP-Emb w/ CLIP ViT-L/14	47.94	80.97	39.00
CLIP-UP-Emb (ours)	59.72	80.30	**51.22**
CLIP-UP-Emb + standard LoRA	59.84	83.35	52.42
CLIP-UP-EmbLoRA (ours)	60.72	83.34	**53.07**

Table 3. Ablation results (%) on LLaVA-1.5-7B.

standard accuracy drop is minimal (*e.g.*, Ovis2-16B), but in others, depending on the application, drops may be too steep. To address this, in App. E we introduce a simple inference-time method for controlling this trade-off.

Open-ended results For open-ended questions, Tabs. 2 and 7 present the RGQA results, and Tab. 8 shows results on TDIUC. On RGQA, CLIP-UP-EmbLoRA surpasses LoRA in almost all cases with gains up to 8.08%. TDIUC shows similar trends, with gains up to 11.32%.

Classification of question type We address the real-world case where inputs may be multiple-choice questions, open-ended questions, or neither. CLIP-UP requires knowing the input type, to determine which embedding to generate. We find that a simple rule-based classifier (described in App. D) is highly effective: it correctly identifies all tested multiple-choice and open-ended inputs, and classifies others (*e.g.*, requests to caption an image) as non-questions. While unnecessary for our data, classification can be done in other ways, such as using the VLM's LLM component.

4.6. Ablation study and limitations

Fig. 4 visualizes the projections learned by CLIP-UP-Emb with t-SNE plots of embedding vectors generated from all samples in the three MM-UPD sub-benchmarks and RGQA test data. As CLIP-UP is intended to discern answerable questions from unanswerable ones, we expect t-SNE to reveal distinct clusters for these two groups. Indeed, clustering appears for all data types: AAD and IVQD clustering is clearest, while IASD clustering is less distinct, likely because the correlation vectors are less suited to capture its textual inconsistency (*i.e.*, question-option mismatches). Open-ended clustering is also less clear, presumably due to the task difficultly (see discussion below).

We perform ablation studies to assess the impact of CLIP-UP's components, with results for CLIP-UP-Emb on MM-UPD shown in Tab. 3. We examine the impact of using Structure-CLIP by replacing the correlation vectors with a constant signal, effectively learning an embedding vector from scratch (line 1 in Tab. 3). This approach is inferior to using Structure-CLIP, confirming that it is a key factor in CLIP-UP and that the gains are not solely due to learning a new embedding vector. Interestingly, the results are not as low as may be expected, presumably because the learned vector encourages the model to extract alignment information already present internally, while the correlation vectors provide additional cues not otherwise accessible.

We examine a CLIP-UP variant where the projection is based on four scalar Structure-CLIP similarities rather than on four correlation vectors (line 2). While this approach performs well, it yields weaker results than CLIP-UP, underscoring the value of the richer information within the correlation vectors. We additionally test CLIP-UP using CLIP ViT-L/14 to generate the correlation vectors instead of Structure-CLIP (line 3), and observe markedly lower results compared to our full CLIP-UP-Emb, underscoring Structure-CLIP's importance. Interestingly, the performance is higher with constant signal (line 1) than with the CLIP ViT-L/14. We hypothesize that the optimization process requires the signal to be highly correlated with the task; otherwise, a conservative solution using the same signal for all inputs may be more effective.

We also test whether CLIP-UP-EmbLoRA benefits from the injection into LoRA layers, or merely from adding LoRA fine-tuning. We observe that it outperforms CLIP-UP-Emb with standard LoRA (lines 5 and 6), highlighting the benefit provided by injecting priors into LoRA.

Finally, while CLIP-UP offers a comprehensive solution for multiple-choice, it may be limited in addressing the diversity of open-ended questions. For instance, it is not expected to help with highly general open-ended prompts (*e.g.*, "What is in this image?"). Another potential limitation stems from the reliance on Structure-CLIP signals, which may be unsuitable in some cases. Please see App. H for further discussion of limitations.

5. Conclusion

This paper introduces CLIP-UP, a lightweight approach for enhancing pre-trained VLMs' ability to withhold responses to unanswerable open-ended and multiple-choice VQA questions. CLIP-UP leverages CLIP-based measures to learn a few linear projections to achieve this, without altering the original VLM weights. Evaluated across several models, CLIP-UP achieves consistent gains over other methods while preserving performance on non-VQA tasks.

Beyond improving models' robustness on unanswerable questions, CLIP-UP also contributes to more trustworthy evaluation of VLMs, as it discourages the strategy of eliminating unlikely options. Thus, it may better uncover VLMs' true knowledge. While this contribution is demonstrated in the vision-language setting, the approach could be similarly applied to other modalities, for example, by leveraging CLAP [61] instead of CLIP in audio-language models.

Acknowledgments

This research was partially supported by Joint NSFC-ISF Research Grant no. 3077/23 and Israel Science Foundation Grant no. 1427/25.

References

[1] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone, 2024. 3, 5, 6

[2] Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision*, pages 73–91, Berlin, Heidelberg, 2025. Springer-Verlag. 3

[3] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. VQA: Visual question answering. In *Proceedings of the IEEE international conference on computer vision*, pages 2425–2433, New York, NY, USA, 2015. IEEE. 1, 3

[4] Omri Avrahami, Kfir Aberman, Ohad Fried, Daniel Cohen-Or, and Dani Lischinski. Break-a-scene: Extracting multiple concepts from a single image. In *SIGGRAPH Asia 2023 Conference Papers*, New York, NY, USA, 2023. Association for Computing Machinery. 3

[5] Zechen Bai, Pichao Wang, Tianjun Xiao, Tong He, Zongbo Han, Zheng Zhang, and Mike Zheng Shou. Hallucination of multimodal large language models: A survey, 2025. 1

[6] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020. 2

[7] Nicola De Cao, Wilker Aziz, and Ivan Titov. Editing factual knowledge in language models, 2021. 3

[8] Khyathi Chandu, Linjie Li, Anas Awadalla, Ximing Lu, Jae Sung Park, Jack Hessel, Lijuan Wang, and Yejin Choi. CertainlyUncertain: A benchmark and metric for multimodal epistemic and aleatoric awareness. In *The Thirteenth International Conference on Learning Representations*, 2025. 2, 3

[9] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling, 2025. 5

[10] Siyuan Cheng, Bozhong Tian, Qingbin Liu, Xi Chen, Yongheng Wang, Huajun Chen, and Ningyu Zhang. Can we edit multimodal large language models?, 2024. 3

[11] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. *See https://vicuna. lmsys. org (accessed 14 April 2023)*, 2(3):6, 2023. 2

[12] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. PaLM: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240): 1–113, 2023. 2

[13] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, and Rongrong Ji. Mme: A comprehensive evaluation benchmark for multimodal large language models, 2024. 3

[14] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion, 2022. 3

[15] Roy Ganz, Yair Kittenplon, Aviad Aberdam, Elad Ben Avraham, Oren Nuriel, Shai Mazor, and Ron Litman. Question aware vision transformer for multimodal reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13861–13871, New York, NY, USA, 2024. IEEE. 3

[16] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6904–6913, New York, NY, USA, 2017. IEEE. 3

[17] Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, et al. HALLUSIONBENCH: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14375–14385, New York, NY, USA, 2024. IEEE. 3

[18] Anisha Gunjal, Jihan Yin, and Erhan Bas. Detecting and preventing hallucinations in large vision language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18135–18143, Washington, DC, USA, 2024. AAAI Press. 3

[19] Yangyang Guo, Fangkai Jiao, Zhiqi Shen, Liqiang Nie, and Mohan Kankanhalli. UNK-VQA: A dataset and a probe into the abstention ability of multi-modal large models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10284–10296, 2024. 3

[20] Danna Gurari, Qing Li, Abigale J Stangl, Anhong Guo, Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P Bigham. VizWiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3608–3617, New York, NY, USA, 2018. IEEE. 1, 3

[21] Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. Aging with GRACE: Lifelong model editing with discrete key-value adaptors. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2023. Curran Associates Inc. 3

[22] Xingwei He, Qianru Zhang, A-Long Jin, Yuan Yuan, and Siu-Ming Yiu. TUBench: Benchmarking large vision-language models on trustworthiness with uuanswerable questions, 2024. 3, 22, 23

[23] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models, 2021. 2, 5, 6, 12

[24] Han Huang, Haitian Zhong, Tao Yu, Qiang Liu, Shu Wu, Liang Wang, and Tieniu Tan. VLKEB: A large vision-language model knowledge editing benchmark, 2024. 3

[25] Yufeng Huang, Jiji Tang, Zhuo Chen, Rongsheng Zhang, Xinfeng Zhang, Weijie Chen, Zeng Zhao, Zhou Zhao, Tangjie Lv, Zhipeng Hu, and Wen Zhang. Structure-CLIP: towards scene graph knowledge to enhance multi-modal structured representations, 2023. 2, 4, 12

[26] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know, 2022. 19

[27] Kushal Kafle and Christopher Kanan. An analysis of visual question answering algorithms. In *Proceedings of the IEEE international conference on computer vision*, pages 1965–1973, 2017. 6, 7, 16, 19, 20

[28] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 12

[29] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213, 2022. 19

[30] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International journal of computer vision*, 123:32–73, 2017. 3

[31] Bohao Li, Yuying Ge, Yixiao Ge, Guangzhi Wang, Rui Wang, Ruimao Zhang, and Ying Shan. SEED-Bench: Benchmarking multimodal large language models. In *CVPR*, pages 13299–13308, 2024. 3, 20

[32] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning*, New York, NY, USA, 2023. JMLR.org. 5

[33] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models, 2023. 1, 3

[34] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft COCO: Common objects in context. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*, pages 740–755, Berlin, Heidelberg, 2014. Springer-Verlag. 12, 13, 16

[35] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems*, pages 34892–34916, Newry, Northern Ireland, 2023. Curran Associates, Inc. 3, 5

[36] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26296–26306, New York, NY, USA, 2024. IEEE. 2, 6

[37] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. LLaVA-NeXT: Improved reasoning, ocr, and world knowledge, 2024. 1, 5, 6

[38] Hanchao Liu, Wenyuan Xue, Yifei Chen, Dapeng Chen, Xiutian Zhao, Ke Wang, Liping Hou, Rongjun Li, and Wei Peng. A survey on hallucination in large vision-language models, 2024. 1, 3

[39] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. MMBench: Is your multi-modal model an all-around player? In *European Conference on Computer Vision*, pages 216–233, Berlin, Heidelberg, 2024. Springer-Verlag. 3, 7

[40] Ilya Loshchilov and Frank Hutter. SGDR: stochastic gradient descent with warm restarts, 2017. 12

[41] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019. 12

[42] Shiyin Lu, Yang Li, Qing-Guo Chen, Zhao Xu, Weihua Luo, Kaifu Zhang, and Han-Jia Ye. Ovis: Structural embedding alignment for multimodal large language model, 2024. 1, 6

[43] Oscar Mañas, Benno Krojer, and Aishwarya Agrawal. Improving automatic VQA evaluation using large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 4171–4179, 2024. 7, 16

[44] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning. Fast model editing at scale, 2022. 3

[45] Atsuyuki Miyai, Jingkang Yang, Jingyang Zhang, Yifei Ming, Qing Yu, Go Irie, Yixuan Li, Hai Li, Ziwei Liu, and Kiyoharu Aizawa. Unsolvable problem detection: Robust understanding evaluation for large multimodal models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. 1, 2, 3, 6, 7, 12, 13, 14, 16, 17, 18, 19, 21, 22, 24, 25, 27

[46] Yulei Niu, Kaihua Tang, Hanwang Zhang, Zhiwu Lu, Xian-Sheng Hua, and Ji-Rong Wen. Counterfactual VQA: A cause-effect look at language bias. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12700–12710, New York, NY, USA, 2021. IEEE. 3

[47] OpenAI. GPT-3.5-turbo-0125, 2023. 16

[48] OpenAI. GPT-4o mini, 2024. 13, 16

[49] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report, 2024. 3

[50] Yusu Qian, Haotian Zhang, Yinfei Yang, and Zhe Gan. How easy is it to fool your multimodal LLMs? an empirical analysis on deceptive prompts, 2024. 2, 3

[51] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763, Cambridge, MA, USA, 2021. PMLR, PMLR. 2, 5, 12

[52] Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning, 2019. 1, 3

[53] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, New York, NY, USA, 2023. IEEE. 3

[54] Dustin Schwenk, Apoorv Khandelwal, Christopher Clark, Kenneth Marino, and Roozbeh Mottaghi. A-OKVQA: A benchmark for visual question answering using world knowledge. In *European conference on computer vision*, pages 146–162, Berlin, Heidelberg, 2022. Springer-Verlag. 13

[55] Oleksii Sidorov, Ronghang Hu, Marcus Rohrbach, and Amanpreet Singh. TextCaps: A dataset for image captioning with reading comprehension. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part II 16*, pages 742–758, Berlin, Heidelberg, 2020. Springer-Verlag. 13, 16

[56] Nick Stracke, Stefan Andreas Baumann, Joshua Susskind, Miguel Angel Bautista, and Björn Ommer. CTRLorALTer: Conditional LoRAdapter for efficient 0-shot control and altering of T2I models. In *European Conference on Computer Vision*, pages 87–103, Berlin, Heidelberg, 2024. Springer-Verlag. 5

[57] Zhiqing Sun, Sheng Shen, Shengcao Cao, Haotian Liu, Chunyuan Li, Yikang Shen, Chuang Gan, Liang-Yan Gui, Yu-Xiong Wang, Yiming Yang, Kurt Keutzer, and Trevor Darrell. Aligning large multimodal models with factually augmented RHLF, 2023. 3

[58] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? exploring the visual shortcomings of multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9568–9578, New York, NY, USA, 2024. IEEE. 3

[59] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and efficient foundation language models, 2023. 2

[60] Yael Vinker, Andrey Voynov, Daniel Cohen-Or, and Ariel Shamir. Concept decomposition for visual exploration and inspiration. *ACM Transactions on Graphics (TOG)*, 42(6): 1–13, 2023. 3

[61] Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, Berlin, Heidelberg, 2023. Springer-Verlag. 8

[62] Yunzhi Yao, Peng Wang, Bozhong Tian, Siyuan Cheng, Zhoubo Li, Shumin Deng, Huajun Chen, and Ningyu Zhang. Editing large language models: Problems, methods, and opportunities, 2023. 3

[63] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. MMMU: A massive multi-discipline multimodal understanding and reasoning benchmark for expert AGI. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, New York, NY, USA, 2024. IEEE. 3

[64] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *The Eleventh International Conference on Learning Representations*, 2023. 21

[65] Ningyu Zhang, Yunzhi Yao, Bozhong Tian, Peng Wang, Shumin Deng, Mengru Wang, Zekun Xi, Shengyu Mao, Jintian Zhang, Yuansheng Ni, et al. A comprehensive study of knowledge editing for large language models, 2024. 3

[66] Yuwei Zhang, Chih-Hui Ho, and Nuno Vasconcelos. Toward unsupervised realistic visual question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15613–15624, New York, NY, USA, 2023. IEEE. 3, 7, 16, 19, 20, 21, 26

[67] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. 1, 3, 5, 6